

Title: Predicting Diabetes Using Support Vector Machines and Health Behavior Data

Introduction

The goal of this project is to investigate and predict the presence of diabetes among adults (age 18 and older) using demographic and behavioral variables. In particular, this analysis focuses on variables such as age, sex, body mass index (BMI), physical activity, and soda consumption to understand how lifestyle and health metrics relate to disease outcomes.

Diabetes is not evenly distributed across populations. According to research, certain demographic groups, including Black, Hispanic, and Native American populations, experience disproportionately higher rates of diabetes and related complications. These disparities are often linked to broader socioeconomic factors such as access to healthcare, education, and resources for disease management (PubMed Central, 2024)

To explore these relationships, this project uses data from the 2022 National Health Interview Survey (NHIS), which contains over 35,000 observations and includes a wide range of demographic, health, and behavioral variables.

Support Vector Machines (SVMs) with linear, radial, and polynomial kernels are used to model and predict diabetes status. The goal is not only to compare model performance, but also to better understand which variables contribute most to predicting disease outcomes.

Theoretical Background

Support Vector Machines (SVMs) are classification models that separate data by finding a boundary that best divides two groups. The model works by maximizing the distance between this boundary and the closest data points from each class, which helps improve how well it performs on new data. Since real-world data is rarely perfectly separable, SVMs use a parameter called C to control how strict the model is. A larger value of C prioritizes correct classification, while a smaller value allows more flexibility and a wider margin. In this project, three types of SVM kernels were used: linear, radial, and polynomial. Each kernel determines the shape of the decision boundary and how the model captures relationships in the data.

The linear SVM assumes that the data can be separated using a straight-line boundary. It is simple, fast, and easy to interpret, making it a good baseline model. However, it may not perform well when the relationship between variables and the outcome is more complex. The main parameter to tune is the cost parameter (C), which controls the tradeoff between margin size and classification accuracy.

The radial (RBF) SVM allows for more flexible, curved decision boundaries by capturing nonlinear relationships between variables. This makes it more powerful when patterns in the data are not linear. In addition to the cost parameter (C), the radial kernel includes a parameter called gamma, which controls how flexible the boundary is. Smaller values of gamma produce smoother boundaries, while larger values allow the model to fit the data more closely. Because of this flexibility, the radial SVM often requires careful tuning to avoid overfitting.

The polynomial SVM models interactions between variables by creating more complex decision boundaries. It introduces an additional parameter called the degree, which controls the complexity of the boundary. While it can capture interactions between features, it is often more sensitive to parameter choices and may not perform as well as the radial kernel in practice.

Model performance was evaluated using accuracy, sensitivity, and specificity. Because the dataset is imbalanced, sensitivity, or the ability to correctly identify individuals with diabetes, is especially important. Class weights were used to improve the model's ability to detect the minority class. Overall, SVMs are useful because they can model both simple and complex relationships, but their performance depends heavily on selecting appropriate parameters and choosing the right kernel for the data.

Methodology

The dataset was cleaned and prepared before building any models. The target variable was created using DIABETICEV and re-coded into a binary outcome, where 1 represents the presence of diabetes and 0 represents no diabetes. Other responses (such as 7, 8, 9, and other placeholder values like 96–999) were treated as missing and removed to avoid introducing noise. The analysis was limited to adults (age ≥ 18) to keep the results consistent and meaningful. A small set of predictors was selected based on both interpretability and data quality, including age, sex, BMI, physical activity, and soda consumption. Extreme values in soda consumption were capped to reduce the effect of outliers, and any remaining missing observations were removed. The data was then split into training and testing sets, categorical variables were converted into dummy variables, and all predictors were scaled to ensure the SVM models performed correctly.

Three SVM models (linear, radial, and polynomial kernels) were trained and compared. Because the dataset is imbalanced, class weights were used so the models would better detect diabetes cases rather than just predicting the majority class. Model performance was evaluated using accuracy, sensitivity, and specificity, with a stronger focus on sensitivity. The radial model was further tuned using cross-validation, testing different values of the cost parameter (0.5, 1, 2, 5) and gamma (0.01, 0.05, 0.1), and selecting the combination that performed best on validation data. The polynomial model was trained on a smaller subset due to computational limitations. Final performance was evaluated on a held-out test set, allowing for a direct comparison of how each kernel performed.

Results

The performance of the three SVM models was compared using accuracy, sensitivity, and specificity, and the results are summarized in a comparison table. The linear model achieved the highest accuracy (0.639), while the radial model achieved the highest sensitivity (0.824), indicating a stronger ability to correctly identify individuals with diabetes. The polynomial model performed the worst overall, with lower values across all metrics.

After tuning, the radial SVM showed a small improvement in accuracy and specificity while maintaining high sensitivity, resulting in the best overall balance between correctly identifying diabetes cases and minimizing false positives. In addition to the model comparison, exploratory plots show that BMI has a clear relationship with diabetes status, while physical activity shows more overlap between

groups. The SVM decision boundary plot further illustrates that individual variables alone do not fully separate the classes, supporting the use of nonlinear models.

Discussion

The results show that predicting diabetes using a small set of demographic and behavioral variables is possible, but not straightforward. While the linear model achieved the highest overall accuracy, this was largely due to the imbalance in the dataset, where most individuals do not have diabetes. In contrast, the radial model performed better in terms of sensitivity, meaning it was more effective at correctly identifying individuals with diabetes. After tuning, the radial model provided the best overall balance between sensitivity and specificity, making it the most useful model in this context.

Looking at the predictors, BMI showed a clear relationship with diabetes, which was supported by the exploratory plots. Individuals with diabetes tended to have higher BMI values on average. Physical activity, on the other hand, showed much more overlap between groups, suggesting that it is a weaker predictor when considered on its own. The decision boundary visualization further supports this, showing that BMI and activity alone are not enough to clearly separate the two classes. This helps explain why more flexible models, like the radial SVM, performed better than simpler linear approaches.

Despite these findings, the models still have limitations. Overall accuracy was lower than the baseline due to the use of class weights, which prioritized detecting diabetes cases over simply predicting the majority class. While this is appropriate for the problem, it also means there is a tradeoff between sensitivity and false positives. In terms of improvements, future work could include additional variables such as diet, income, or access to healthcare, which are known to influence diabetes risk. Other modeling approaches could also be explored, or more extensive tuning could be applied to further improve performance. Overall, the results highlight both the potential and the limitations of using SVMs for predicting health outcomes.

Conclusions

This project demonstrates that support vector machines can be used to predict diabetes using a combination of demographic and behavioral variables. While no single variable perfectly predicts disease status, factors such as BMI play an important role, and nonlinear models like the radial SVM are better suited to capture these relationships.

From a broader perspective, these findings highlight the importance of understanding how lifestyle and health behaviors contribute to disease risk. Although the models developed in this analysis are not intended for clinical use, they can help identify patterns and trends that may be useful for public health efforts. With more comprehensive data and additional predictors, similar models could be used to better identify at-risk populations and support early intervention strategies.

Sources

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